Emmanuel Oladokun

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Summary

Oxford DPhil graduate with expertise developing **generative models** (diffusion, flow-matching) for high-dimensional spatio-temporal data, with first-author publications and industry collaborations. Experienced in **improving model performance through data-centric methods**, including synthetic data, evaluation, and iterative experimentation on large, noisy datasets. Worked with **fine-tuning multi-modal models (e.g. CLIP)** and leveraging agents/LLMs to accelerate workflows. Interested in building and deploying **scalable AI systems** that perform reliably in the real world.

Education

DPhil Engineering Science CDT, University of Oxford (period includes internship) Oct 2020 – March 2026

Since Oct 2021, in collaboration with GE HealthCare:

- Developed and applied **deep generative models** to create synthetic datasets for downstream tasks.
- Designed models **for controlled generation conditioned on structured inputs**; improved downstream performance by up to **10%** via targeted augmentation, rebalancing, counterfactuals, and **iterative evaluation.**
- Implemented a **custom JVP FlashAttention processor** for one-step latent video generation, enabling a **50× reduction in inference latency** and **3× reduction in GPU memory usage**.

MEng Engineering Science, University of Oxford 2016 – 2020

Modules include: **Machine Learning, Software Engineering**, Optimisation, **Deep Learning**, Computer Vision

Research & Industry Experience

Alan Turing Intern, The Alan Turing Institute / GCHQ Jan 2024 – Jul 2024

- Curated and processed **large-scale image datasets (∼ 300k samples)** and developed pipelines for training and evaluating **SOTA** generative models.
- Researched, implemented, and trained models for **data generation**, with a focus on **generalisability.**
- Built and ran scalable training workflows using **AWS Cloud** services (**S3**, **SageMaker**, **CloudWatch**) for data storage, model training, and experiment monitoring.

Developer & Supervisor, University of Oxford / GE HealthCare Oct 2020 – Jun 2022

- Co-developed and maintained an **open-source tool** for automated validation of DICOM images, improving **data quality** within clinical ML pipelines while ensuring privacy compliance. ECIQC
- **Led technical design and implementation using both C++ and Python**, including **CI/CD** pipelines and automated tests to ensure reliability as the codebase evolved.
- **Supervised 5 PhD students**, mentoring on software development practices and contributing to key project deliverables.

Master's Thesis Research, University of Oxford Oct 2019 – Jun 2020

- **Used Python** and MATLAB to develop **mathematical** and **statistical models** for the circadian blood pressure rhythm.
- Managed and performed statistical analysis on **noisy data** consisting of **4M records** from **208,948** hospital admissions.
- Completed an independent research project under **time constraints**, achieving a **First Class Grade**.

Software Engineering Intern, Schlumberger Jul 2019 – Sep 2019

- Engineered **TensorFlow** and **Scikit-Learn** ML models for a production reservoir simulator.
- Achieved a **30–50%** reduction in computational costs; findings contributed to a scientific **publication**.

First-author Publications

- EchoLVFM: One-Step Video Generation via Latent Flow Matching for Echocardiogram Synthesis; (2026); EchoLVFM
  Early Accepted (top 9% of 4,601); Flow Matching; **One-Step Inference**; Video Generation; GenAI; ,
- From Transthoracic to Transesophageal: Cross-Modality Generation using LoRA Diffusion; MICCAI-ASMUS (2025).
  Best Paper Award; Diffusion Models; Multi-Modal Learning; **Parameter-Efficient Fine-Tuning**; , , Proprietary
- Transesophageal Echocardiography Generation using Anatomical Models; MICCAI-DALI (2023).
  Anatomical Mesh Priors; Contrastive Learning; Synthetic Data; , Proprietary
- Machine-Learning Informed Prediction of Linear Solver Tolerance for Non-Linear Solution Methods in Numerical Simulation. ECMOR XVII (2020) Random Forest Regression; Linear Algebra; , Proprietary

Skills

Programming Languages: Python, MATLAB, and **C++**

Libraries: **HuggingFace**, Accelerate; **PyTorch**, DDP, Torchvision; NumPy, SciPy.

Cloud & Deployment: aws **AWS**, Vercel.

Methods: Diffusion models, latent video generation, flow matching, **generative modelling**, **multi-modal learning**, transformer architectures, **vision-language models (e.g., CLIP)**, synthetic data generation and augmentation.

Awards

Selected Awards

- **Best Paper (Sep 2025):** Awarded **'Best Paper'** at ASMUS-MICCAI 2025 for work on diffusion-based image generation using data from multiple domains.
- **OxAI Hackathon (Mar 2026):** Co-developed a transfer learning prototype to estimate visceral fat proxies from facial imagery; built an end-to-end multimodal training pipeline in six hours, earning **3rd place**. Face2Visceral
- **Google HQ Hackathon (Oct 2019):** Co-developed an ML application, delivering a working **prototype in less than 24 hours** and earning **2nd place** out of 12 teams. HandRiGhT
- **UKMT Maths Challenge (Nov 2015):** Awarded a Silver Certificate.

Leadership and Service

Engineering Tutor, St Peter's College — 2022

- Delivered tutorials in Mathematics and Physics to undergraduate engineering students, and assessed problem sheets.

Volunteer, THRIVE — Nov 2018 – Dec 2022

- Led weekly sports sessions for **50+ children** from deprived communities, mentoring participants while fostering teamwork, discipline, and leadership.

Alumni Relations Officer, St Peter's College, University of Oxford — Nov 2016 – Jun 2018

- Established strong ties as an ambassador through connecting undergraduates with alumni, and networking with professionals.
- Raised **£5,862** for the college during a telethon and learned protocols for dealing with sensitive and confidential information.

Interests

Drumming

Active drummer performing monthly in live services to congregations of up to **200** people.